

Target-site mutation accumulation among ALS inhibitor-resistant Palmer amaranth

Shilpa Singh,^a Vijay Singh,^b Reiofeli A Salas-Perez,^a Muthukumar V Bagavathiannan,^b Amy Lawton-Rauh^c and Nilda Roma-Burgos^{a*}

Abstract

BACKGROUND: Palmer amaranth (*Amaranthus palmeri* S. Wats) is one of the most common and troublesome weeds in the USA. Palmer amaranth resistance to acetolactate synthase (ALS) inhibitors is widespread in the USA, as in Arkansas. The cross-resistance patterns and mechanism of resistance are not known. Experiments were conducted to determine cross-resistance to ALS inhibitors and identify target-site mutations in 20 Palmer amaranth localities from 13 counties in Arkansas.

RESULTS: All Palmer amaranth localities tested had plants cross-resistant to imazethapyr, flumetsulam, primisulfuron, pyriothiac and trifloxysulfuron. The dose of trifloxysulfuron that caused 50% control was 21–56-fold greater for resistant accessions than for susceptible ones. All but three resistant plants analyzed had one or two relative copies of ALS; one plant had seven relative copies. All resistant plants tested (18 localities) carried the Trp574Leu mutation, which is known to confer broad resistance to ALS inhibitors, supporting the cross-resistance pattern observed. Besides the Trp574Leu mutation, 30% of localities had individuals with one additional resistance-conferring mutation including Ala122Thr, Pro197Ala or Ser653Asn.

CONCLUSION: The Trp574Leu mutation in ALS is the primary mechanism of resistance to ALS inhibitors in Palmer amaranth from Arkansas, USA. In some localities, multiple mutations have accumulated in one plant. All localities tested contained plants with resistance to five families of ALS inhibitors. Localities with extremely high resistance to ALS inhibitors, and those outside the subset we studied, may harbor non-target site resistance mechanisms. ALS inhibitors are generally no longer effective on Palmer amaranth in these localities from the US mid-south.

© 2018 Society of Chemical Industry

Keywords: TSR; ALS-inhibitor resistance; multiple mutations; broad cross-resistance; double mutations; ALS copy number

1 INTRODUCTION

Acetolactate synthase (ALS)-inhibiting herbicides were commercialized in 1982 and this was a milestone in weed control history. These herbicides are used in very small quantities (as low as 4 g ai ha⁻¹), compared with the high doses of many other herbicides, and this reduced the total herbicide load in the environment.¹ Many herbicides in this group have broad-spectrum efficacy, soil residual activity, wide application windows, high margins of crop safety and low mammalian toxicities.² ALS is the first common enzyme in the biosynthetic pathway of the branched-chain amino acids valine, leucine and isoleucine.³ Herbicides belonging to five chemical families inhibit ALS: sulfonylureas (SUs), imidazolinones (IMIs), pyrimidinylthiobenzoates (PTBs), triazolopyrimidines (TPs) and sulfonaminocarbonyl triazolinones (SACTs).⁴ Among the ALS inhibitors traditionally used for Palmer amaranth (*Amaranthus palmeri* S. Wats.) control are imazaquin and imazethapyr (IMIs) in soybean. Pyriothiac (PTB) and trifloxysulfuron (SU) have been used also for supplemental control in cotton, although by themselves, these herbicides are relatively weak on Palmer amaranth. Long-term use of these chemistries has resulted in the evolution of resistance across many weed species. Since the first case of ALS herbicide resistance in prickly lettuce (*Lactuca serriola* L.) in 1986,⁵

160 weed species (monocots and dicots) have been reported to be resistant to ALS-inhibiting herbicides.⁴

Collectively, *Amaranthus* species have the most cases of resistance to various herbicide modes of action because of their high genetic diversity (predominantly outcrossing), high seed production (large population size) and continuous emergence (high propensity to escape).^{6–8} *Amaranthus* species that have evolved resistance to ALS-inhibiting herbicides include Palmer amaranth,⁹ Powell amaranth (*Amaranthus powellii* S. Watson),¹⁰ common waterhemp (*Amaranthus rudis* (Moq.) Sauer),¹¹ livid amaranth (*Amaranthus lividus* L.),¹² redroot pigweed (*Amaranthus retroflexus* L.),¹⁰ smooth pigweed (*Amaranthus hybridus* L.),¹³ tall waterhemp

* Correspondence to: N Roma-Burgos, Department of Crop, Soil and Environmental Sciences, University of Arkansas, Fayetteville, AR 72701, USA. E-mail: nburgos@uark.edu

a Department of Genetics and Biochemistry, Clemson University, Clemson, SC, USA

b Department of Soil and Crop Sciences, Texas A&M University, College Station, TX, USA

c University of Arkansas Cooperative Extension Service, Lonoke, AR, USA

(*Amaranthus tuberculatus* (Moq.) Sauer),¹⁴ spiny amaranth (*Amaranthus spinosus* L.)¹⁵ and prostrate pigweed (*Amaranthus blitoides* L.).¹⁶ Resistance to ALS inhibitors in Palmer amaranth occurs in at least 13 states in the USA.⁴

Mechanisms of resistance to ALS inhibitors are: (1) target site modifications;¹⁷ (2) processes (structural, biochemical or physiological) external to the target site that result in a reduced amount of herbicide reaching the site of action; or (3) protection from the phytotoxic effects of the herbicide.^{18,19} Mechanisms 2 and 3 are collectively known as non-target site resistance (NTSR), among which are reduced absorption or translocation, increased herbicide detoxification, sequestration of herbicide in the vacuoles or cell walls, and increased production of antioxidants.

The majority of ALS-resistant weed species harbor target site resistance (TSR) mechanisms. Mutations at one of eight ALS loci (Ala122, Pro197, Ala205, Asp376, Arg377, Trp574, Ser653 and Gly654) are known to confer resistance to ALS-inhibiting herbicides.^{20,21} Cross-resistance associated with an altered ALS occurs among these herbicide classes: (1) SU and TP, (2) IMI and PTB, (3) IMI and SU or (4) resistance to all classes.²² Amino acid substitutions at Ala122 or Ser653 confer resistance to IMI herbicides with low-level resistance to SUs.^{23,24} Pro197 substitution confers resistance to SUs,²⁵ but low risk of cross-resistance to IMIs. Asp376Glu substitution confers resistance to all five chemical families of ALS inhibitors,²⁶ and substitution at Trp574 confers resistance to at least the IMIs, SUs and TPs.

In tall waterhemp and smooth pigweed, ALS mutations at positions Trp574 or Ser653,²⁷ and Ala122, Ala205, Asp376, Trp574 or Ser653, respectively²⁶ confer resistance to ALS inhibitors. Resistance to ALS inhibitors in Palmer amaranth in the USA is due to amino acid substitutions at Ala122, Pro197, Trp574 or Ser653.^{15,28,29} Palmer amaranth localities across the USA are also mostly resistant to glyphosate, due to multiplication of the target gene, 5-enolpyruvylshikimate-3-phosphate synthase (*EPSPS*).³⁰ Whether target gene amplification contributes to some cases of ALS inhibitor resistance in Palmer amaranth is not known. The objectives of this study were to identify known resistance-conferring mutations in the ALS gene of Palmer amaranth and to determine whether ALS gene amplification also occurs in some of these ALS inhibitor-resistant Palmer amaranth localities. This information will help predict which classes of ALS inhibitors may still be useful for managing Palmer amaranth.

2 MATERIALS AND METHODS

2.1 Plant materials and general testing for resistance to ALS herbicides

Mature inflorescences of Palmer amaranth were collected from 20 fields representing 13 counties in Arkansas between 2008 and 2011. Each field was sampled once. At least 10 plants were sampled per field. Samples were air-dried, threshed manually and a composite seed sample was prepared by mixing equal amounts of seed per plant. This composite seed sample is referred to as an accession representing a field locality. Composite seed from each accession was planted in 50-cell trays (Redwayfeed Garden and Pet Supply, Redway, CA, USA) filled with Sunshine® potting medium (Sunshine premix #1®, Sun Gro Horticulture, Bellevue, WA, USA). The experiment was set up as randomized complete block design with two replications and two runs (one 50-cell tray per replication) with one seedling per cell. Thus, a total of 200 plants (5–7 cm tall) representing a field locality were treated with recommended doses of imazethapyr (IMI) at 71 g ha⁻¹

(Pursuit®, BASF, Research Triangle Park, NC, USA), flumetsulam (TP) at 69 g ha⁻¹ (Python®, Dow AgroSciences, Indianapolis, IN, USA), primisulfuron-methyl (SU) at 53 g ha⁻¹ (Beacon®, Syngenta Crop Protection, Inc., Research Triangle Park, NC, USA), pyriithiobac (PTB) at 73 g ha⁻¹ (Staple®, DuPont Crop Protection, Wilmington, DE, USA) and trifloxysulfuron (SU) 8 g ha⁻¹ (Envoke®, Syngenta Crop Protection, Research Triangle Park, NC, USA). A tank-mixture of pyriithiobac and trifloxysulfuron was also applied with 0.25% v/v NIS. Plants were treated in a spray chamber using a motorized boom fitted with two flat-fan nozzles (TeeJet spray nozzles, Spraying Systems Co., Wheaton, IL, USA) delivering 187 L ha⁻¹ at 32 psi. At 21 days after treatment (DAT), each plant was evaluated visually for injury relative to the non-treated control on a scale of 0–100%, where 0 was no injury and 100% was dead. The data were analyzed, by herbicide, using analysis of variance (ANOVA) in JMP Pro v. 12 (SAS Institute Inc., Cary, NC, USA). Hierarchical clustering of accessions was conducted using injury and mortality data. Hierarchical cluster analysis was performed based on Ward's minimum variance method in JMP Pro v. 12 (SAS Institute). This method creates cluster groupings by minimizing the within-group sum of squares. The number of clusters was selected based on fit statistic, with the largest Cubic Clustering Criterion (CCC) value.

The primisulfuron data were analyzed separately by run because plants in each run were sprayed at different sizes (run 1 at 5–8 cm tall and run 2 at < 5 cm tall). Smaller plants were sprayed in the second run to minimize false positives, although a wide range of plant sizes is present in farmers' fields at the time of herbicide application. The difference in plant size affected the frequency of survivors and the level of injury observed.

2.2 Evaluation of resistance level to trifloxysulfuron

A dose–response assay was conducted on 20 Palmer amaranth accessions showing differential responses to trifloxysulfuron. Seeds were planted in 11 × 11 cm² pots filled with Sunshine Mix LC1 potting soil (Sun Gro Horticulture Canada Ltd., Vancouver, BC, Canada). Each experimental unit (pot) contained five seedlings. The resistant accessions were sprayed with eight doses of trifloxysulfuron at 0, 1, 2, 4, 8, 16, 32 and 64 g ha⁻¹ (recommended field rate = 8 g ha⁻¹). The susceptible (SS) accession was sprayed with 0, 0.5, 1, 2, 4, 8 and 16 g ha⁻¹. Herbicides were applied as described above. The experiment was conducted in a randomized complete block design with four replications. The number of survivors and injury per plant were recorded at 21 DAT. Data were analyzed using SigmaPlot v. 13 (Systat Software, Inc., San Jose, CA, USA). The data were fitted with a three-parameter log-logistic model (Eqn 1) to predict the trifloxysulfuron dose that would cause 50 or 90% injury:

$$Y = c / [1 + e^{-a(x-b)}] \quad (1)$$

where Y is the % injury, a is the asymptote, b is the slope, c is the inflection point and x is the trifloxysulfuron dose.

2.3 ALS gene copy number determination

Leaf tissues from four survivor plants per accession were collected at 21 DAT of trifloxysulfuron (8 g ha⁻¹). For susceptible accessions, tissues were collected from the non-treated plants. Leaf tissues were frozen immediately in liquid nitrogen and stored at –80 °C until processed. Genomic DNA was extracted from 100 mg of leaf tissue using a modified CTAB protocol,³¹ quantified using a NanoDrop spectrophotometer (Thermo Scientific, Wilmington, DE, USA), and checked for quality by gel electrophoresis.

A quantitative real-time polymerase chain reaction (qPCR) was used to determine the *ALS* gene copy number relative to *A36*. *A36* (*Celosia trigyna* PI649298) is a RNA dead-box helicase. The *A36* primers were designed using Biologn and Primer 3 software from sequences of the *Amaranthus* genus: *A36_F244* (5'-TTGGAAGTGTGAGCAACC-3') and *A36_R363* (5'-GAACCACTTCCACCAAAAC-3').³² To amplify the *ALS* gene, the primer sets *ALS_F2* (5'-AGCTCTGGAACGTGAAGGC-3') and *ALS_R2* (5'-TCAATTAACCGTCCGGG-3') designed by Gaines *et al.*³⁰ were used. For the qPCR, 25- μ L reactions were prepared using 12.5 μ L of Bio-Rad iQ SYBR Green Supermix, 1 μ L of the forward and reverse primers (10 μ M) and 10 ng of gDNA. The thermoprofile used consisted of 15 min denaturation at 95 °C, 40 cycles of 95 °C for 30 s, and 60 °C for 60 s. This program was followed by a melt-curve analysis of 81 cycles of 55 °C for 30 s.³³ A negative control consisting of primers with no template DNA was included. No amplification products were observed in negative control reactions. Data were analyzed using a modification of the $2^{-\Delta\Delta C_t}$ method to express genomic copy number of *ALS* relative to *A36* as $\Delta C_t = (C_t, A36 - C_t, ALS)$, and relative increase in genomic *ALS* copy number was expressed as $2\Delta C_t$.³⁴ Each locality was represented by four biological replicates and each reaction had three technical replicates (for each primer pair).

2.4 *ALS* gene sequence analysis

Genomic DNA was used to amplify the complete sequence of the *ALS* gene. The PCR consisted of 25 μ L 2 $\times$ PCR master mix (Takara Bio USA, Inc.), 2 μ L of both the forward and reverse primers (10 μ M), 4 μ L gDNA (50 ng μ L⁻¹) and 17 μ L of water to make a total volume of 50 μ L. The following primers

were used to sequence the full gene (2 kb): forward primer *PAALS_F*-5'-ATGGCGTCCACTTCAACAAAC-3', reverse primer *PAALS_R*-5'-GGTGATGGAAGAAGGGCTTATTAG-3'; and internal primers *PAALS_F2*-5'-AGGATATTCCTAGAATTGTTAAGG-3', *PAALS_F3*-5'-ATGCGGTTGTAAGTACCGGTGT-3', *PAALS_R1*-5'-CTGGACCTGTTTGTATTGATA-3'. The PCR was performed with the following conditions: initial denaturation at 94 °C for 5 min, followed by 30 cycles of denaturation at 94 °C for 1 min, annealing at 66 °C for 1.5 min, extension at 72 °C for 2 min and a final extension at 72 °C for 5 min. The PCR tubes were held at 4 °C until processed. The PCR product was visualized on a 1% agarose gel to confirm the expected fragment size. The PCR product was purified using a NucleoSpin® Gel and PCR Clean-up kit (Takara Bio USA) and was sequenced at the Institute for Plant Genomics and Biotechnology (IPGB, Texas A&M University, College Station, TX, USA). The nucleotide sequences were aligned using Bioedit software.³⁵ Sequences were aligned with *Arabidopsis thaliana* *ALS* sequence (NM_114714.3) and a susceptible Palmer amaranth sequence (KT833339.1) available at GenBank.

3 RESULTS AND DISCUSSION

3.1 Response of Palmer amaranth to ALS-inhibiting herbicides

The Palmer amaranth accessions evaluated in this study were resistant to five ALS-inhibiting herbicides (imazethapyr, flumetsulam, primisulfuron-methyl, pyriithiobac-sodium, trifloxysulfuron) and the mixture of pyriithiobac plus trifloxysulfuron (Fig. 1 and Table 1). The accessions differentiated into three clusters (Fig. 1) based on mortality and the level of injury on the survivors. For

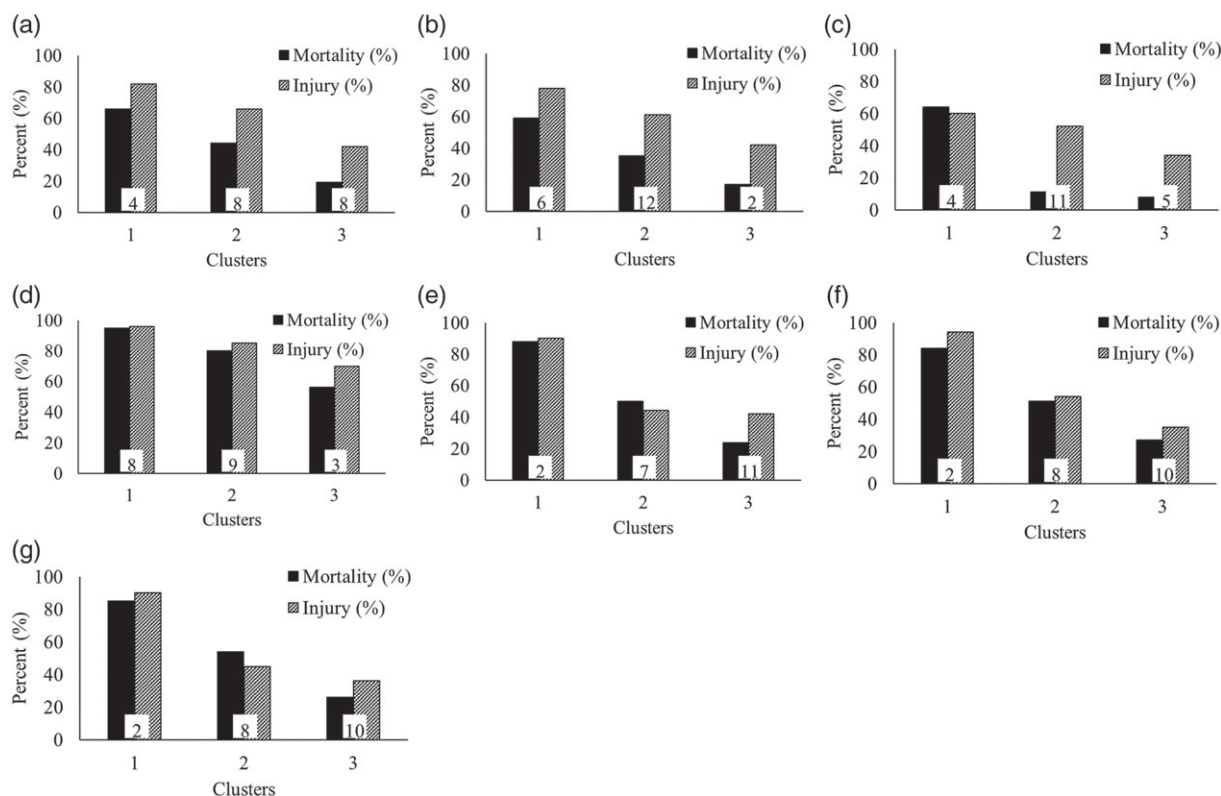


Figure 1. Graphical representation of the cluster analysis of 20 *Amaranthus palmeri* accessions from Arkansas, USA, treated with ALS inhibitors: (a) imazethapyr, (b) flumetsulam, (c,d) primisulfuron-methyl, (e) pyriithiobac-sodium, (f) trifloxysulfuron, and (g) pyriithiobac plus trifloxysulfuron. For primisulfuron, each run was analyzed separately due to plant size differences across runs. The number of accessions per cluster is shown as boxed text.

Table 1. Response of Palmer amaranth accessions to imazethapyr, flumetsulam, primisulfuron-methyl, pyriithiobac-sodium and trifloxysulfuron, Arkansas, USA

Accession ^a	Imazethapyr ^b		Flumetsulam		Primisulfuron				Pyriithiobac		Trifloxy-sulfuron		Pyriithiobac + Trifloxysulfuron	
	INJ ^c	MOR ^c	INJ	MOR	INJ, R1 ^d	MOR, R1	INJ, R2 ^d	MOR, R2	INJ	MOR	INJ	MOR	INJ	MOR
	----- % -----													
CRA08-A	85	69	80	79	57	9	96	95	40	48	62	55	50	46
JAC08-B	32	13	65	36	44	6	82	80	38	51	58	37	47	51
LEE08-A	68	50	60	40	58	15	83	74	45	25	30	30	39	55
MIS08-B	64	41	53	31	49	12	82	74	25	15	45	22	30	38
PHI08-A	46	14	60	41	52	18	76	75	35	20	20	47	28	40
POI08-A	28	13	57	32	45	6	71	62	40	30	35	32	55	21
STF08-A	69	57	76	48	56	9	98	95	55	12	29	38	38	15
CON09-A	85	69	80	75	64	80	94	96	80	75	95	90	80	70
CLA11-A	46	27	66	39	39	5	88	87	42	18	46	35	52	58
JAC11-A	60	55	73	38	47	61	80	89	53	56	58	45	40	49
JAC11-B	47	26	52	70	52	6	97	95	47	46	61	31	40	47
LAW11-A	47	16	64	34	50	19	87	84	50	28	60	70	46	58
MIS11-A	63	35	39	13	38	5	62	47	35	30	45	18	31	27
MIS11-B	63	46	54	31	46	14	86	81	29	38	32	25	45	21
MIS11-C	58	38	44	21	38	2	76	59	46	22	51	12	38	33
MIS11-D	67	44	53	32	47	7	89	78	40	43	40	62	50	51
POI11-B	40	15	62	42	25	25	80	73	55	36	30	21	48	64
PRA11-B	74	39	81	71	54	3	97	96	49	60	50	74	39	65
WHI11-A	82	69	72	50	65	20	85	92	38	42	35	20	28	18
SS ^e	85	90	92	95	89	97	95	96	100	100	100	100	100	100

^a Accessions were selected to represent different levels of resistance to ALS-inhibitor herbicides.

^b Plants were sprayed with imazethapyr (71 g ha⁻¹), flumetsulam (69 g ha⁻¹), primisulfuron-methyl (53 g ha⁻¹), pyriithiobac-sodium (73 g ha⁻¹), trifloxysulfuron (8 g ha⁻¹), and a tank mix of pyriithiobac (73 g ha⁻¹) plus trifloxysulfuron (8 g ha⁻¹) at 5–7 cm tall. Treatments were applied with 0.25% v/v non-ionic surfactant (NIS).

^c INJ (% injury) and MOR (% mortality) were assessed 21 days after herbicide application.

^d R1 = Run 1 and R2 = Run 2 for primisulfuron. Plants were 5–8 cm tall in Run 1 and < 5 cm tall in Run 2.

^e Susceptible standard.

imazethapyr, flumetsulam, pyriithiobac, and trifloxysulfuron the first cluster consisted of accessions with 88% mortality, and the survivors incurred 90% injury at 21 DAT. The two susceptible accessions (SS and CON09-A) were in this cluster. This cluster also contained two and four accessions, respectively, that had < 80% injury after treatment with imazethapyr or flumetsulam. Accessions in the second cluster incurred 66%, 61%, 44%, and 54% injury, respectively, with the four herbicides. Several survivors were barely affected (15% injury). This group was classified as resistant. The third cluster had average injury levels ranging from 35% to 42%, which put this group into the highly resistant category.

For primisulfuron, the plants in run 2, which were smaller when treated, incurred more injury and had higher mortality than the larger plants in run 1 (Fig. 1 and Table 1). Even with the small plants in run 2, accessions MIS11-A, MIS11-C and POI08-A incurred only 56% mortality; thus, were classified as highly resistant. The ALS herbicides, imazethapyr (for soybean and other legumes), flumetsulam (for corn and soybean), primisulfuron (for corn), pyriithiobac (for cotton) and trifloxysulfuron (for cotton) are used to control several broadleaved weed species. Pyriithiobac applied pre or post controls *Amaranthus* species.³⁶ Similarly, trifloxysulfuron applied post controls many weeds including smooth pigweed and Palmer amaranth.³⁷ These once-effective herbicides now generally fail to control Palmer amaranth. Applying a mixture of pyriithiobac and trifloxysulfuron resulted in the same level of control across

all accessions as when these herbicides were applied separately (Fig. 1 and Table 1). Failure to control Palmer amaranth with the tank-mixture confirmed cross-resistance to both herbicide families (PTB and SU). Cross-resistance to these families of ALS inhibitors was strong.

As observed with other ALS-inhibitor herbicides, the accessions separated into three clusters in response to pyriithiobac plus trifloxysulfuron (Fig. 1). The two accessions in the first cluster were susceptible with 85% mortality and 90% injury, 21 DAT. The eight accessions in the second cluster were resistant, with an average of 46% survivors sustaining 45% injury. The highly resistant cluster (cluster 3) consisted of 10 accessions that had only 26% mortality. The most resistant survivors (with 0–10% injury) were from Clay, Mississippi, Phillips, St. Francis and White Counties (Table 1).

These localities reflected the widely reported occurrence of ALS-inhibitor resistance in Palmer amaranth across the USA.⁴ Resistance was reported in Kansas in 1993, Arkansas and Tennessee in 1994, North Carolina in 1995, South Carolina in 1997, Georgia in 2000, Florida and Mississippi in 2008, Arizona in 2012, Illinois in 2013, and Delaware and Maryland in 2014.⁴ Cross-resistance to multiple ALS-inhibiting herbicides is common in Palmer amaranth. In 2001, imazaquin-resistant Palmer amaranth localities from Arkansas were reported to be cross-resistant to chlorimuron (SU), diclosulam (TP) and pyriithiobac (PTB).³⁸

Table 2. Nonlinear regression parameters and trifloxysulfuron dose required for 50% control of ALS-susceptible and ALS-resistant Palmer amaranth accessions from Arkansas, USA

Accession	Regression equation	R ²	LL ^a	ED ₅₀ ^b	UL ^c	R/S ^d
----- g ha ⁻¹ -----						
CRA08-A	$Y = 53/[1 + e^{-0.207(x-7.878)}]$	0.95	9	22	35	28
JAC08-B	$Y = 53/[1 + e^{-0.139(x-9.981)}]$	0.94	13	30	47	38
LEE08-A	$Y = 63/[1 + e^{-0.094(x-15.460)}]$	0.94	21	30	40	38
MIS08-B	$Y = 60/[1 + e^{-0.184(x-8.168)}]$	0.92	12	17	23	22
PHI08-A	$Y = 59/[1 + e^{-0.102(x-12.261)}]$	0.94	18	29	40	37
POI08-A	$Y = 55/[1 + e^{-0.133(x-10.342)}]$	0.94	15	27	39	35
STF08-A	$Y = 61/[1 + e^{-0.148(x-10.577)}]$	0.93	14	21	28	26
CON09-A	$Y = 100/[1 + e^{-0.397(x-2.391)}]$	0.99	2	2	3	3
CLA11-A	$Y = 52/[1 + e^{-0.094(x-11.758)}]$	0.96	7	44	81	56
JAC11-A	$Y = 55/[1 + e^{-0.171(x-9.045)}]$	0.94	13	23	33	29
JAC11-B	$Y = 75/[1 + e^{-0.270(x-5.180)}]$	0.95	11	19	27	24
LAW11-A	$Y = 56/[1 + e^{-0.130(x-9.321)}]$	0.96	15	26	37	33
MIS11-A	$Y = 50/[1 + e^{-0.189(x-8.312)}]$	0.94	18	32	102	41
MIS11-B	$Y = 57/[1 + e^{-0.079(x-14.143)}]$	0.97	22	39	56	50
MIS11-C	$Y = 60/[1 + e^{-0.084(x-13.757)}]$	0.95	21	35	48	44
MIS11-D	$Y = 52/[1 + e^{-0.192(x-8.147)}]$	0.94	8	24	40	31
POI11-B	$Y = 60/[1 + e^{-0.073(x-17.236)}]$	0.94	25	39	53	50
PRA11-B	$Y = 54/[1 + e^{-0.171(x-8.870)}]$	0.94	13	22	31	28
WHI11-A	$Y = 56/[1 + e^{-0.127(x-11.787)}]$	0.89	17	29	41	37
SS ^e	$Y = 101/[1 + e^{-0.794(x-1.341)}]$	0.99	0.6	0.8	1.0	1.0

^a LL = lower limit of 95% confidence interval.

^b ED₅₀ (effective dose causing 50% injury) was estimated using a non-linear logistic three-parameter regression equation: $Y = c/[1 + e^{-a(x-b)}]$ where Y is % injury, a is the asymptote, b is the slope, c is the inflection point and x is the trifloxysulfuron dose.

^c UL = upper limit of 95% confidence interval.

^d Resistance levels (R/S) were calculated as the ratio of the ED₅₀ value for the putative resistant accession relative to the susceptible standard.

^e Susceptible standard.

In Georgia, imazapic-resistant accessions were also resistant to chlorimuron, diclosulam and pyriithobac.³⁹ In the current study, cross-resistance to all the ALS-inhibitor herbicide families in Palmer amaranth is common. The first ALS-inhibitor cross-resistance study in Arkansas included seven localities in Lawrence County³⁸; this current study covered 13 counties.

3.2 Resistance level to trifloxysulfuron

The recommended field rate (8 g ha⁻¹) of trifloxysulfuron controlled the susceptible accessions completely (Table 2). It would take 0.80 and 2 g ha⁻¹ trifloxysulfuron to kill 50% of SS and CON09-A accessions, respectively. The 18 other accessions had 22–56-fold resistance relative to the SS standard. The accessions from Clay, Mississippi, Phillips and White Counties had high resistance to trifloxysulfuron. ED₉₀ values could be calculated only for SS (3 g ha⁻¹) and CON09-A (8 g ha⁻¹). The remaining accessions were highly resistant to trifloxysulfuron and could not be controlled 90% by the highest dose applied. A high level of resistance to ALS inhibitors had been reported before in several *Amaranthus* species. Palmer amaranth localities with 2800-fold resistance to imazethapyr have been reported.⁴⁰ In Israel, a prostrate pigweed accession had 790-fold resistance to ALS inhibitors.¹⁶ Some smooth pigweed accessions exhibited up to 537-fold resistance to imazethapyr, but with less resistance to chlorimuron (15-fold).⁴¹ The mechanisms of resistance was not investigated in these studies; therefore, the reason for extreme resistance to ALS inhibitors is not understood.

3.3 ALS genomic copy number relative to A36

Gene amplification, as a mechanism of resistance, is rare in plant species. Recently, *EPSPS* gene amplification has been found as the primary mechanism of resistance in some glyphosate-resistant weed species including *Amaranthus* spp., *L. perenne* ssp. *multiflorum*, and *K. scoparia*. In most studies of *EPSPS* gene amplification, *ALS* has been used as a reference gene because it is generally expressed as a single copy. However, selection pressure with ALS inhibitors may likewise isolate genotypes with increased *ALS* gene copies, which could result in a resistant phenotype. This was not the case for the Palmer amaranth we studied. Palmer amaranth in these localities, whether susceptible or resistant to ALS inhibitors, had one or two copies of *ALS* (Fig. 2) with a few exceptions. At least one plant each from MIS08-B, MIS11-A and WHI11-A had more *ALS* copies. A maximum of seven copies was observed in a single plant from Mississippi County and four copies in a plant from White County. These plants were highly resistant, with < 30% injury, but the increase in copy number was too small for it to have contributed to the resistance level. A single plant each from MIS08-B, MIS11-A and WHI11-A that had more *ALS* copies harbored double mutations involving Ser653Asn, Pro197Ala and Ala122Thr mutations, respectively, in addition to the Trp574Leu mutation. All other ALS-resistant plants had only the Trp574Leu mutation. Should there be more resistant plants with increased *ALS* gene copies in the future, gene amplification should be verified using southern blotting or fluorescence *in situ* hybridization as done in the case of *EPSPS*.^{30,42} Recently in Japan, one accession of shortawn foxtail (*Alopecurus aequalis* var. *sonomensis*) resistant to thifensulfuron-methyl was found to

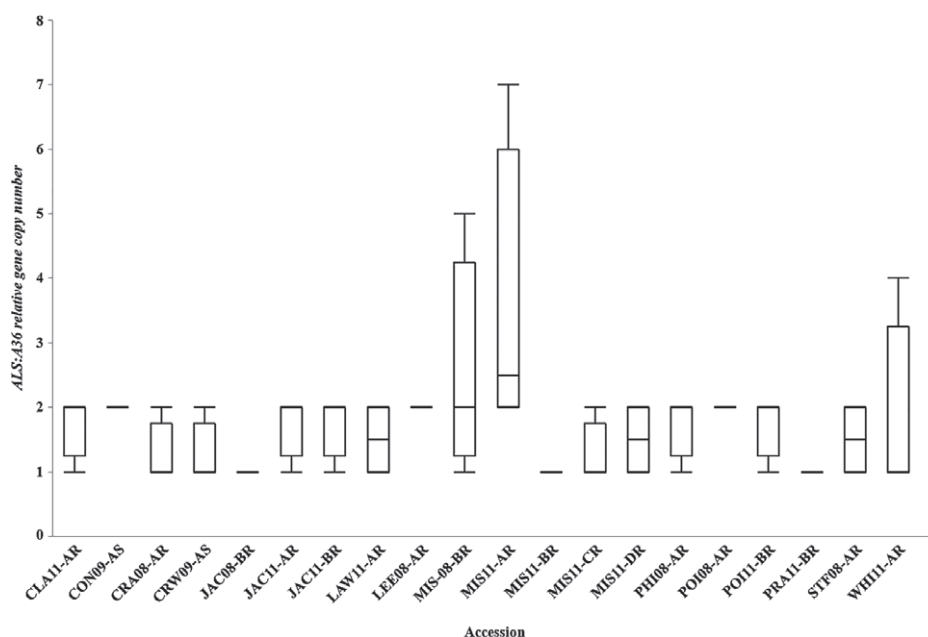


Figure 2. Relative *ALS:A36* gene copy number among susceptible (S) and resistant (R) Palmer amaranth accessions (four plants per accession). Box plot shows median values (horizontal line inside the box), first and third quartile values (box outlines), and minimum and maximum values (whiskers).

Accession	Level of resistance to ALS herbicides						Mutations			
	Imaz	Flumet	Primi	Pyri	Trifloxy	Pyri+ Trifloxy	Ala122Thr	Pro197Ala	Ser653Asn	Trp574Leu
CRA08-A	MR	MR	S	R	R	HR				
JAC08-B	HR	R	R	R	R	HR				
LEE08-A	R	R	R	HR	HR	HR				
MIS08-B	R	R	R	HR	HR	R				
PHI08-A	HR	R	R	HR	HR	R				
POI08-A	HR	R	HR	HR	HR	R				
STF08-A	R	R	MR	HR	HR	R				
CON09-A	S	S	S	S	S	S				
CLA11-A	HR	R	R	HR	R	HR				
JAC11-A	R	R	R	R	R	HR				
JAC11-B	HR	R	S	R	R	HR				
LAW11-A	HR	R	R	HR	R	HR				
MIS11-A	R	HR	HR	HR	HR	R				
MIS11-B	R	R	R	HR	HR	R				
MIS11-C	R	HR	HR	HR	HR	R				
MIS11-D	R	R	R	R	R	HR				
POI11-B	HR	R	R	HR	HR	HR				
PRA11-B	R	MR	S	R	R	HR				
WHI11-A	MR	R	R	R	HR	R				
SS	S	S	S	S	S	S				

Figure 3. Graphic presentation of resistance-conferring *ALS* mutations in *ALS*-resistant Palmer amaranth from Arkansas, USA and the corresponding population-level resistance (S, susceptible; MR, moderately resistant; R, resistant; HR, highly resistant) to imazethapyr (Imaz), flumetsulam (Flumet), primisulfuron-methyl (Primi), pyriithiobac-sodium (Pyri), and trifloxysulfuron (Trifloxy).

have up to four copies of *ALS*.⁴³ The presence of multiple copies was confirmed by cloning the *ALS* gene. However, resistance to thifensulfuron-methyl in these accessions was attributed to *ALS* mutations (substitution of Pro197 with Ser, Leu or Thr). In the case of glyphosate target, *EPSPS*, it takes ~ 10 copies before a resistant phenotype was observed.^{30,44} The *ALS*-resistant Palmer amaranth we analyzed had < 10 copies. Collectively, these suggest that the slight elevation in relative *ALS* gene copy in some *ALS*-resistant Palmer amaranth plants most likely does not contribute to the resistance level.

3.4 *ALS* sequence analysis

Of the 20 Palmer amaranth accessions, 18 were resistant to *ALS* herbicides representing four families: IMIs (imazethapyr), PTBs

(pyriithiobac), SUs (primisulfuron-methyl; trifloxysulfuron) and TPs (flumetsulam). In Arkansas, Palmer amaranth resistance to *ALS* inhibitors was first reported in a locality from Lawrence County in 1994; the mechanism of resistance was hypothesized to be due to an altered target site based on reduced sensitivity of the *ALS* enzyme to *ALS* herbicides.³⁸ Mutation at the target site is the primary mechanism of resistance to *ALS* inhibitors reported thus far. The sequence alignment of *ALS* gene from 20 Palmer amaranth accessions (four plants per accession) revealed that all resistant plants contained a Trp574Leu mutation (Figs 3 and 4). This point mutation is known to confer resistance to IMIs, SUs and TPs in various weed species.⁴ This explains the broad cross-resistance to all *ALS*-inhibitor herbicide families among these accessions.

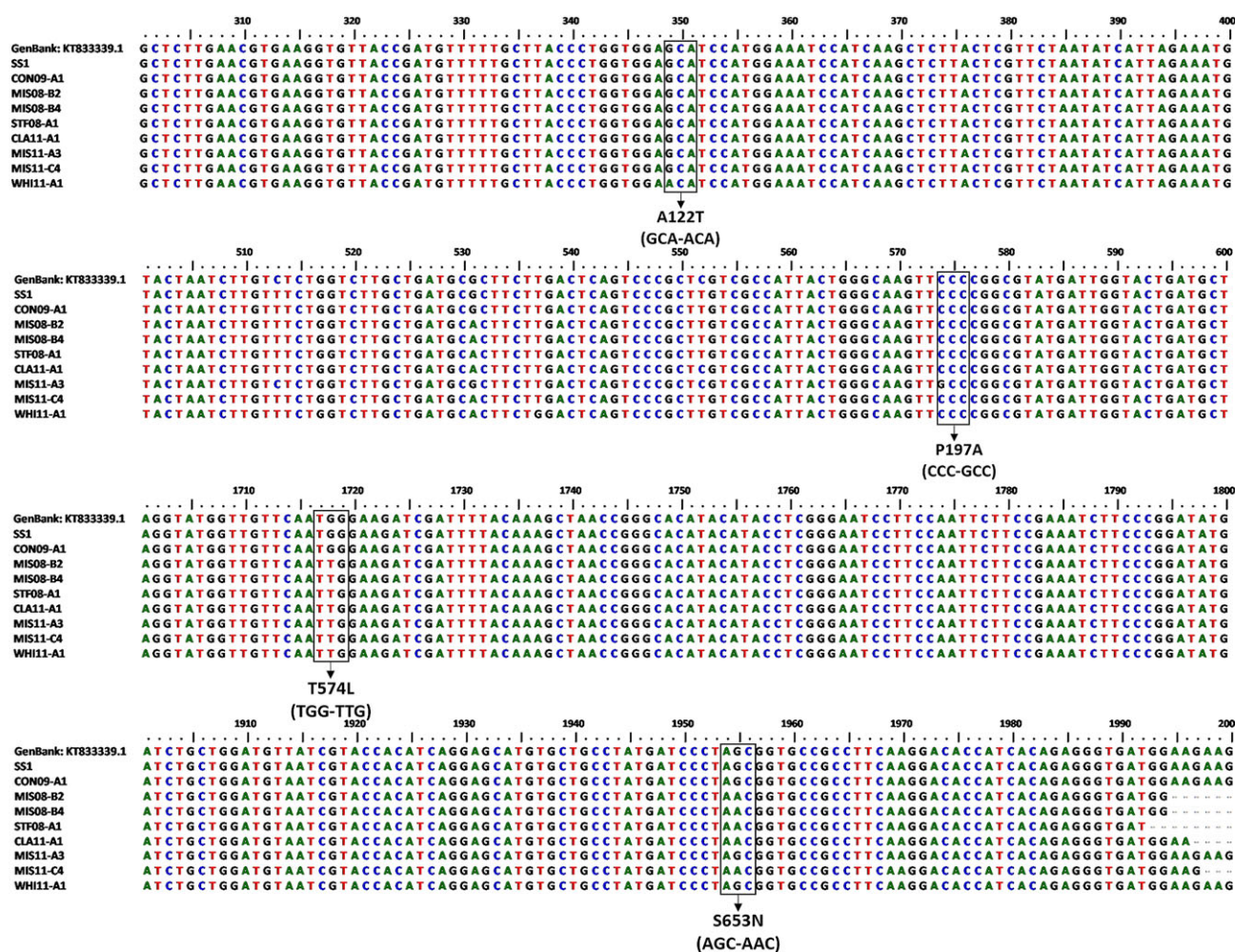


Figure 4. Sequence alignment of a region of *ALS* gene from susceptible (SS1, CON09-A1, and GenBank: KT833339.1) and representative *ALS*-resistant Palmer amaranth (MIS08-B2-B4, STF08-A1, CLA11-A1, MIS11-A3, MIS11-C4 and WHI11-A1) from Arkansas, USA. The sequences presented here show the site of four resistance-conferring *ALS* mutations. Amino acid numbering is based on *Arabidopsis thaliana* *ALS* sequence.

The Trp574Leu mutation has also been detected in other *Amaranthus* species. A prostrate pigweed population from Israel that had Trp574Leu mutation was also cross-resistant to IMIs, PTBs, SUs and TPs.¹⁶ In Illinois, a common waterhemp population that was cross-resistant to IMIs, SUs and TPs harbored Trp574Leu mutation.⁴⁵ In Canada, redroot pigweed and Powell amaranth harboring Trp574Leu substitution were cross-resistant to imazethapyr and thifensulfuron-methyl.¹⁰

Of the 80 Palmer amaranth plants for which *ALS* was sequenced, five (from CLA11-A, MIS08-B, MIS11-C and STF08-A counties) also had a guanine to adenine substitution at position 653 resulting in a Ser653Asn mutation (Fig. 3). The Ser653Asn mutation is known to confer resistance only to IMIs.²² Plants that carried both the Trp574Leu and Ser653Asn mutations were resistant to all *ALS*-inhibitor herbicides tested (Table 1). These mutations were reported recently in localities from the adjacent state, Mississippi where Trp574Leu was found more frequently than Ser653Asn¹⁵ as among the Arkansas localities. The resistant Palmer amaranth accessions from Mississippi were cross-resistant to IMIs, PTBs and SUs. The *ALS*-resistant Palmer amaranth discovered recently in Brazil all carried the Trp574Leu mutation, whereas 42% of these plants also had the Ser653Asn mutation.⁴⁶ The Palmer amaranth from Brazil was resistant to SUs, but was not tested for

cross-resistance to other *ALS* herbicide families. In tall waterhemp, Trp574Leu endowed high levels of resistance to the *ALS* inhibitors tested including IMIs, SUs and TPs, whereas Ser653Asn caused resistance only to the IMIs.²⁷ Researchers across the globe do not test the same suite of *ALS*-inhibitor herbicide families and most tests do not include a representative of all families. The data collectively show that the Trp574Leu mutation in *Amaranthus* *ALS* confers resistance to all families of *ALS* inhibitors.

One of four plants from MIS11-A had Pro197Ala mutation in addition to Trp574Leu. This plant was highly resistant. The MIS11-A accession was resistant to both primisulfuron and trifloxysulfuron, incurring 67% and 45% injury, respectively. The presence of both Pro197Ala and Trp574Leu mutations in one plant could endow a high level of resistance to a population. To verify this, more rigorous population genetic model testing, verifying genetic connectivity of localities into populations as well as the movement of alleles, needs to be performed using whole genome sequence data. Models of molecular evolution also need to be performed. Such analyses could determine the frequency of individuals harboring these *ALS* mutations and determine whether these mutations are rare or embedded with other modes of adaptation. Recently in Kansas, Pro197Ser mutation was found in Palmer amaranth localities that showed high resistance to chlorsulfuron

(SU).²⁹ No other ALS resistance-conferring mutations were found in the Kansas localities.

One of four plants from the WH11-A accession harbored Ala122Thr in addition to the Trp574Leu mutation. The Ala122Thr mutation endows high resistance to IMLs but low resistance to SUs; whereas Pro197Ala confers resistance to SUs.²² In Argentina, two ALS inhibitor-resistant Palmer amaranth localities had the mutations Trp574Leu, Ser653Asn and Ala122Ser. One locality had the Trp574Leu mutation only, whereas the other had both Ser653Asn and Ala122Ser mutations.²⁸

The Pro197Ala and Ala122Thr mutations in ALS had not been reported previously in Palmer amaranth. To verify whether these mutations occur in the same allele (double mutation) or in different alleles, the amplified gene product will need to be cloned before sequencing. *De novo* genetic changes can be either hereditary or somatic. The accumulation of non-deleterious ALS mutations in one plant is expected to occur in outcrossing species, especially dioecious ones like Palmer amaranth, *via* gene flow. Resistance to ALS inhibitors is inherited as an incompletely dominant trait.²² In this case, the outcrosses will harbor two different resistant alleles, one from each resistant parent. On the other hand, multiple mutations can occur in the same gene in one plant *via* somatic changes caused by an error in DNA replication or by environmental stress. Herbicide application selects for an individual, or individuals, harboring a *de novo* mutation in the target site that allows the plant to survive the herbicide action. The accumulation of multiple mutations in the same ALS allele can occur in this manner if selection continues in a population that has already been preselected for resistance to the same herbicide mode of action. In this study, 30% of the accessions contained individuals with two known ALS-resistance conferring mutations and the cross-resistance patterns conformed to the expected phenotypes conferred by these mutations.

Because Palmer amaranth has evolved resistance to several herbicide sites of action, knowledge of the mechanisms of herbicide resistance is useful for planning sustainable weed management practices. The consequences of having multiple target-site mutations in many *Amaranthus* species is detrimental to chemical weed management. Palmer amaranth can hybridize with other *Amaranthus* species and this spreads the resistance problem further. For example, an ALS-resistant hybrid between Palmer amaranth and spiny amaranth was found to carry the Trp574Leu mutation, which originally existed in Palmer amaranth.¹⁵ Spiny amaranth is not a problematic weed in crop fields, but is a pernicious weed in pastures.

In conclusion, this study presents a comprehensive assessment of the resistance status to ALS inhibitors and the prevailing resistance mechanisms in the state of Arkansas. Resistance to ALS inhibitors in Palmer amaranth is widespread (95% of localities) in the state. ALS inhibitor-resistant localities are cross-resistant to IMLs, PTBs, SUs and TPs. Resistance is primarily due to TSR mechanism involving the Trp574Ser mutation, with some cases of two mutant alleles involving either Ala122Thr, Pro197Ala or Ser653Asn. Based on accumulated knowledge in the literature, these additional mutations could broaden cross-resistance and increase the resistance level. The current ALS inhibitors used in agronomic crops could no longer control Palmer amaranth in these localities.

ACKNOWLEDGEMENTS

The authors thank Marlon Bastiani, Tiago Kaspary, Josiane Carla Argenta and Pamela Carvalho de Lima for their assistance in greenhouse bioassays. This research was funded by the Arkansas Soybean Research and Promotion Board and Syngenta Crop Protection Inc.

REFERENCES

- Bellinder RR, Gummeson G and Karlsson C, Percentage driven government mandates for pesticide reduction: the Swedish model. *Weed Technol* **8**:350–359 (1994).
- Mazur BJ and Falco SC, The development of herbicide resistant crops. *Annu Rev Plant Physiol Mol Biol* **40**:441–470 (1989).
- Durner J, Gailus V and Böger P, New aspects on inhibition of plant acetolactate synthase by chlorsulfuron and imazaquin. *Plant Physiol* **95**:1144–1149 (1991).
- Heap I, *The International Survey of Herbicide Resistant Weeds*. [Online]. Available: www.weedscience.org/ [10 March 2018].
- Mallory-Smith CA, Thill DC and Dial MJ, Identification of sulfonylurea herbicide-resistant prickly lettuce (*Lactuca serriola*). *Weed Technol* **4**:163–168 (1990).
- Lovell ST, Wax LM, Horak MJ and Peterson DE, Imidazolinone and sulfonylurea resistance in a biotype of common waterhemp (*Amaranthus rudis*). *Weed Sci* **1**:789–794 (1996).
- Chan KF and Sun M, Genetic diversity and relationships detected by isozyme and RAPD analysis of crop and wild species of *Amaranthus*. *Theor Appl Genet* **95**:865–873 (1997).
- Norsworthy JK, Griffith GM, Scott RC, Smith KL and Oliver LR, Confirmation and control of glyphosate-resistant Palmer amaranth (*Amaranthus palmeri*) in Arkansas. *Weed Technol* **22**:108–113 (2008).
- Gaeddert JW, Peterson DE and Horak MJ, Control and cross-resistance of an acetolactate synthase inhibitor-resistant Palmer amaranth (*Amaranthus palmeri*) biotype. *Weed Technol* **11**:132–137 (1997).
- McNaughton KE, Letarte J, Lee EA and Tardif FJ, Mutations in ALS confer herbicide resistance in redroot pigweed (*Amaranthus retroflexus*) and Powell amaranth (*Amaranthus powellii*). *Weed Sci* **53**:17–22 (2005).
- Hinz JR and Owen MD, Acetolactate synthase resistance in a common waterhemp (*Amaranthus rudis*) population. *Weed Technol* **11**:13–18 (1997).
- Manley BS, Wilson HP and Hines TE, Smooth pigweed (*Amaranthus hybridus*) and livid amaranth (*A. lividus*) response to several imidazolinone and sulfonylurea herbicides. *Weed Technol* **10**:835–841 (1996).
- Manley BS, Wilson HP and Hines TE, Characterization of imidazolinone-resistant smooth pigweed (*Amaranthus hybridus*). *Weed Technol* **1**:575–584 (1998).
- Patzoldt WL, Tranel PJ and Hager AG, Variable herbicide responses among Illinois waterhemp (*Amaranthus rudis* and *A. tuberculatus*) populations. *Crop Prot* **21**:707–712 (2002).
- Molin WT, Nandula VK, Wright AA and Bond JA, Transfer and expression of ALS inhibitor resistance from Palmer amaranth (*Amaranthus palmeri*) to an *A. spinosus* × *A. palmeri* hybrid. *Weed Sci* **64**:240–247 (2016).
- Sibony M and Rubin B, Molecular basis for multiple resistance to acetolactate synthase-inhibiting herbicides and atrazine in *Amaranthus blitoides* (prostrate pigweed). *Planta* **216**:1022–1027 (2003).
- Saari LL, Cotterman JC and Thill DC, Resistance to acetolactate synthase inhibiting herbicides, in *Herbicide Resistance in Plants: Biology and Biochemistry*, ed. by Powles SB and Holtum JAM. Lewis, Ann Arbor, MI, pp. 83–139 (1994).
- Menendez JM, De Prado R, and Devine MD, Chlorsulfuron cross-resistance in a chlorotoluron-resistant biotype of *Alopecurus myosuroides*, in *Proceedings of Brighton Crop Protection Conference, Weeds*. The British Crop Protection Council, Farnham, UK, p. 319 (1997).
- Veldhuis LJ, Hall LM, O'Donovan JT, Dyer W and Hall JC, Metabolism-based resistance of a wild mustard (*Sinapis arvensis* L.) biotype to ethametsulfuron-methyl. *J Agric Food Chem* **48**:2986–2990 (2000).
- Yu Q and Powles SB, Resistance to AHAS inhibitor herbicides: current understanding. *Pest Manag Sci* **70**:1340–1350 (2014).

- 21 Tranel PJ, Wright TR and Heap IM, *Mutations in Herbicide-Resistant Weeds to ALS Inhibitors*. Available: Error! Hyperlink reference not valid [15 March 2018].
- 22 Powles SB and Yu Q, Evolution in action: plants resistant to herbicides. *Annu Rev Plant Biol* **61**:317–347 (2010).
- 23 Bernasconi P, Woodworth AR, Rosen BA, Subramanian MV and Siehl DL, A naturally occurring point mutation confers broad range tolerance to herbicides that target acetolactate synthase. *J Biol Chem* **270**:17381–17385 (1995).
- 24 Devine MD and Eberlein CV, Physiological, biochemical and molecular aspects of herbicide resistance based on altered target sites, in *Herbicide Activity: Toxicology, Biochemistry and Molecular Biology*, ed. by Roe RM, Burton JD and Kuhr RJ. IOS, Amsterdam, pp. 159–185 (1997).
- 25 Guttieri MJ, Eberlein CV, Mallory-Smith CA, Thill DC and Hoffman DL, DNA sequence variation in domain A of the acetolactate synthase gene of herbicide resistant and susceptible weed biotypes. *Weed Sci* **40**:670–676 (1992).
- 26 Whaley CM, Wilson HP and Westwood JH, A new mutation in plant ALS confers resistance to five classes of ALS-inhibiting herbicides. *Weed Sci* **55**:83–90 (2007).
- 27 Patzoldt WL and Tranel PJ, Multiple ALS mutations confer herbicide resistance in waterhemp (*Amaranthus tuberculatus*). *Weed Sci* **55**:421–428 (2007).
- 28 Larran AS, Palmieri VE, Perotti VE, Lieber L, Tuesca D and Permingeat HR, Target-site resistance to acetolactate synthase (ALS)-inhibiting herbicides in *Amaranthus palmeri* from Argentina. *Pest Manag Sci* **73**:2578–2584 (2017).
- 29 Nakka S, Thompson CR, Peterson DE and Jugulam M, Target site-based and non-target site based resistance to ALS inhibitors in Palmer amaranth (*Amaranthus palmeri*). *Weed Sci* **65**:681–689 (2017).
- 30 Gaines TA, Zhang W, Wang D, Bukun B, Chisholm ST, Shaner DL *et al.*, Gene amplification confers glyphosate resistance in *Amaranthus palmeri*. *Proc Natl Acad Sci* **107**: 1029–1034 (2010).
- 31 Doyle JJ and Doyle JS, A rapid DNA isolation procedure for small quantities of fresh leaf tissue. *Phytochem Bull* **19**:11–15 (1990).
- 32 Singh S, Singh V, Lawton-Rauh A, Bagavathiannan MV and Roma-Burgos N, *EPSPS* gene amplification primarily confers glyphosate resistance among Arkansas Palmer amaranth (*Amaranthus palmeri*) populations. *Weed Sci* **66**:293–300 (2018).
- 33 Chandi A, Milla-Lewis SR, Giacomini D, Westra P, Preston C, Jordan DL *et al.*, Inheritance of evolved glyphosate resistance in a North Carolina Palmer amaranth (*Amaranthus palmeri*) biotype. *Int J Agron* **2012**:1–7 (2012).
- 34 Gaines TA, Shaner DL, Ward SM, Leach JE, Preston C and Westra P, Mechanism of resistance of evolved glyphosate-resistant Palmer amaranth (*Amaranthus palmeri*). *J Agric Food Chem* **59**:5886–5889 (2011).
- 35 Hall TA, BioEdit: a user-friendly biological sequence alignment editor and analysis program for windows 95/98/NT. *Nucleic Acids Symp Ser* **41**:95–98 (1999).
- 36 Dotray PA, Keeling JW, Henniger CG and Abernathy JR, Palmer amaranth (*Amaranthus palmeri*) and devil's claw (*Proboscidea louisianica*) control in cotton (*Gossypium hirsutum*) with pyriithobac. *Weed Technol* **10**:7–12 (1996).
- 37 Porterfield D, Wilcut JW and Askew SD, Weed management with trifloxysulfuron, fluometuron, and prometryn in cotton. *Weed Sci* **50**:642–647 (2002).
- 38 Burgos NR, Kuk YI and Talbert RE, *Amaranthus palmeri* resistance and differential tolerance of *Amaranthus palmeri* and *Amaranthus hybridus* to ALS-inhibitor herbicides. *Pest Manag Sci* **57**:449–457 (2001).
- 39 Wise AM, Grey TL, Prostko EP, Vencill WK and Webster TM, Establishing the geographical distribution and level of acetolactate synthase resistance of Palmer amaranth (*Amaranthus palmeri*) accessions in Georgia. *Weed Technol* **23**:214–220 (2009).
- 40 Sprague CL, Stoller EW, Wax LM, Horak MJ and Palmer amaranth, (*Amaranthus palmeri*) and common waterhemp (*Amaranthus rudis*) resistance to selected ALS-inhibiting herbicides. *Weed Sci* **45**:192–197 (1997).
- 41 Whaley CM, Wilson HP and Westwood JH, ALS resistance in several smooth pigweed (*Amaranthus hybridus*) biotypes. *Weed Sci* **54**:828–832 (2006).
- 42 Dillon AJ, Varanasi VK, Danilova T, Koo DH, Nakka S, Peterson D *et al.*, Physical mapping of amplified 5-enolpyruvylshikimate-3-phosphate synthase gene copies in glyphosate-resistant *Amaranthus tuberculatus*. *Plant Physiol* **12**:01427 (2016).
- 43 Iwakami S, Shimono Y, Manabe Y, Endo M, Shibaike H, Uchino A *et al.*, Copy number variation in acetolactate synthase genes of thifensulfuron-methyl resistant *Alopecurus aequalis* (shortawn fox-tail) accessions in Japan. *Front Plant Sci* **8**:254 (2017).
- 44 Salas RA, Dayan FE, Pan Z, Watson SB, Dickson JW, Scott RC *et al.*, *EPSPS* gene amplification in glyphosate-resistant Italian ryegrass (*Lolium perenne* ssp. *multiflorum*) from Arkansas. *Pest Manag Sci* **68**:1223–1230 (2012).
- 45 Foes MJ, Tranel PJ, Wax LM and Stoller EW, A biotype of common waterhemp (*Amaranthus rudis*) resistant to triazine and ALS herbicides. *Weed Sci* **1**:514–520 (1998).
- 46 Küpper A, Borgato EA, Patterson EL, Netto AG, Nicolai M, Carvalho SJ *et al.*, Multiple resistance to glyphosate and acetolactate synthase inhibitors in Palmer Amaranth (*Amaranthus palmeri*) identified in Brazil. *Weed Sci* **65**:317–326 (2017).